

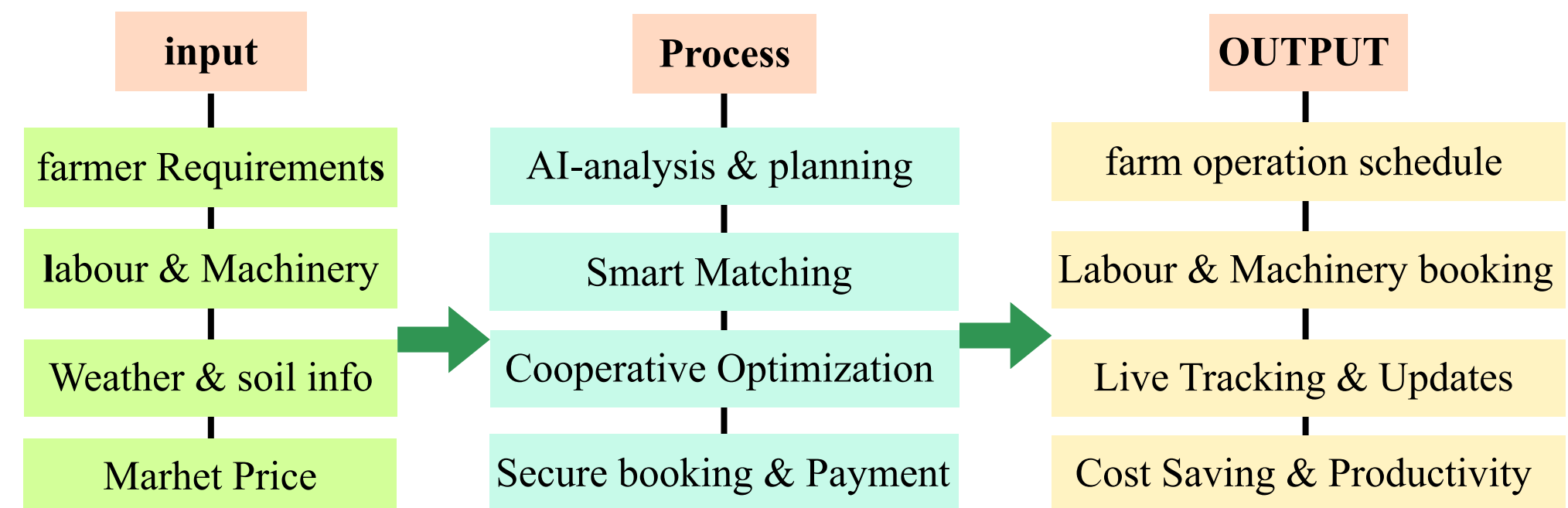
- **Problem Statement ID – SHK007**
- **Problem Statement Title-** AI smart farm operations & Resource Coordination platform
- **Theme-** Smart Agriculture & Rural Digital Transformation
- **PS Category-** Software
- **Team ID-PAR5ZV**
- **Team Name-Team KrishiX**

PROPOSED SOLUTION

- An AI-powered **Smart Farm Operations Platform** that **connects Farmers, Labourers, and Machinery Owners** in a single digital ecosystem.
- Farmers can **search, compare**, and instantly **book verified labour** and **machinery** based on real-time availability, pricing, and ratings
- The AI Farm Operations Planner generates a **personalized crop-wise schedule** and sends timely reminders for every farming activity.
- Smart **Cooperative Booking** enables nearby farmers to share machinery, reducing transportation costs, fuel consumption, and waiting time.
- Live **GPS tracking, OTP verification**, secure **digital payments**, and **Smart Trust Score** ensure transparency, safety, and reliability.
- A **centralized dashboard** allows users to manage bookings, monitor resources, track earnings, and view analytics in one place.
- The platform helps ensure timely farm operations, better resource utilization, lower farming costs, and **increased productivity**.

KEY USP's

- Generates a **personalized crop-wise farming schedule** with timely reminders.
- **Instantly matches verified labour and machinery** based on real-time availability, pricing, and ratings.
- **Enables nearby farmers to share machinery**, reducing transportation costs and increasing utilization.
- OTP verification, Smart Trust Score, and secure digital **payments ensure transparency** and reliability.
- **Manage bookings, live tracking, earnings, analytics**, and resources from a **single platform**.



IMPACT AND BENEFITS

BENIFITS

- Improves **farm productivity** through AI-based planning and timely operations.
- Reduces **labour shortages** by connecting farmers with verified labourers instantly.
- Optimizes **machinery utilization** using Smart Cooperative Booking.
- Lowers **transportation costs** by recommending nearby resources.
- Promotes **digital farming** with online booking, tracking, and secure payments.
- Builds **trust and transparency** through OTP verification, ratings, and Smart Trust Score.
- Supports **sustainable agriculture** by reducing fuel usage and resource wastage.
- Strengthens **rural economy** by creating employment and increasing farmers' income.

IMPACT

- **AI-based** farm planning and timely reminders
- Easy **labour & machinery booking** with real-time tracking
- Lower **farming costs** and improved productivity
- More **booking opportunities** and steady income
- Better **machinery utilization** with less idle time
- **Secure payments** and transparent transaction
- Efficient **sharing of local resources**
- **Sustainable** and collaborative **farming** ecosystem

EXPECTED OUTCOMES

Higher Farm
Productivity

Better Resource
Utilization

Faster Digital
Booking

Increased
Farmer
Income

Transparent
Operations

Sustainable
Agriculture

FEASIBILITY AND VIABILITY

Feasibility

- **Android & Web platform** with a simple and user-friendly interface.
- Integrates farmers, labourers, machinery owners & digital payments **on one platform**.
- Real-time **booking, live tracking, notifications & multilingual support**.
- Built using **cost-effective** cloud technologies, making deployment affordable.
- **Easily scalable** to new villages, districts and states.

Viability

- **Solves** the labour shortage and machinery availability problem.
- **Reduces** waiting time, transportation cost and fuel consumption.
- **Improves** farm productivity through AI-based planning.
- **Creates** new earning opportunities for labourers and machinery owners.
- **Generates** revenue through service/booking fees and premium services.
- **Supports** future integration with government schemes, weather APIs and smart agriculture services.

challenges	our solution
Labour shortage	AI matches nearby verified labourers instantly
Machinery unavailability	Real-time machinery booking with live availability
High transportation cost	Smart Cooperative Booking reduces travel cost
rust & payment issues	OTP verification, ratings & Smart Trust Score
Lack of transparency	Centralized dashboard with tracking & analytics
Limited digital adoption	Simple multilingual interface with voice assistance

RESEARCH INSIGHTS

- Studied the challenges faced by **farmers, labourers, and machinery owners** in rural Maharashtra.
- Identified major issues such as **labour shortages, machinery underutilization**, and **delayed farm operations**.
- Explored **AI-based farm planning, cooperative resource sharing, and digital booking systems** as practical solutions.
- Reviewed national initiatives promoting **Digital Agriculture** and rural digital transformation.
- These findings formed the foundation of the **Smart Kopergaon Platform**.

REFERENCES

Indian Council of Agricultural Research (ICAR)
<https://icar.org.in>

Ministry of Agriculture & Farmers Welfare
<https://agricoop.gov.in>

Department of Agriculture, Maharashtra
<https://krishi.maharashtra.gov.in>

e-NAM – National Agriculture Market
<https://enam.gov.in>

Food and Agriculture Organization (FAO)
<https://www.fao.org>

PROTOTYPE LINK

TECHNICAL APPROACH



TechnicalStack

Frontend

Backend

Flowchart

Mobile Web App

Prototype